

Unit 7, Lesson 4: Factoring a Trinomial With a Leading Coefficient Greater Than 1



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can use algebra tiles to factor a trinomial with a leading coefficient greater than 1.
- ☐ I can factor a trinomial with a leading coefficient greater than 1.



7.4.1 Warm-Up: Growth Mindset

1. Each statement describes the celebrity indicated.

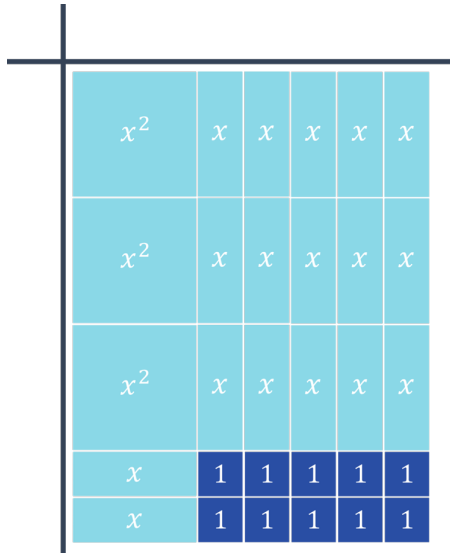
Statement	Celebrity
He was never offered a top Division 1 scholarship. Scouting Reports said his explosiveness and athleticism were below standard.	Steph Curry
His first book was rejected by 27 different publishers. They said his material was too different and would never sell.	Dr. Seuss
He was fired from a newspaper for not being creative enough. They said he lacked imagination and had no good ideas.	Walt Disney

- a. How did a growth mindset help these celebrities?
- b. What can you do to maintain a growth mindset if you get frustrated with today's (or any day's) lesson to help you keep a growth mindset?



7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$

1. Use the algebra tiles to determine the two binomials that are factors of $3x^2 + 17x + 10$.



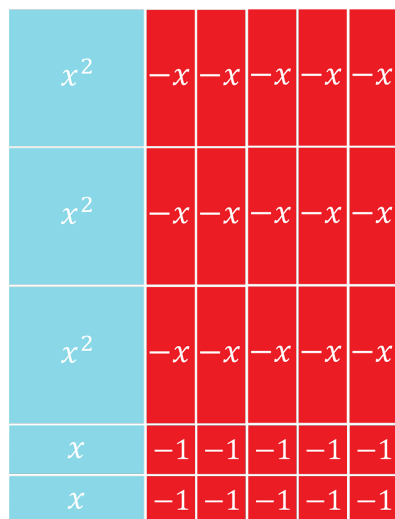
2. Discuss with your partner any differences you notice between the binomial factors in question 1 and factoring trinomials where $a = 1$. Summarize your discussion.

3. The steps for applying the distributive property to multiply $(3x + 2)(x - 5)$ are shown.

Step	$(3x + 2)(x - 5)$	Result
Distribute $3x$ to x	$(3x + 2)(x - 5)$ 	$3x^2$
Distribute $3x$ to -5	$(3x + 2)(x - 5)$ 	$-15x$
Distribute 2 to x	$(3x + 2)(x - 5)$ 	$2x$
Distribute 2 to -5	$(3x + 2)(x - 5)$ 	-10

- a. Based on the results in the table, what is the product of $(3x + 2)(x - 5)$?

- b. The algebra tiles diagram models the result of $(3x + 2)(x - 5)$.



Draw a rectangle around $-15x$ in the algebra tiles diagram for $3x^2 - 13x - 10$.

- c. Discuss with your partner how the algebra tiles diagram demonstrates the distribution of $3x$ to -5 . Summarize your discussion.

4. Compare the algebra tiles diagram for $3x^2 + 17x + 10$ to the algebra tiles diagram for $3x^2 - 13x - 10$. Describe the differences.

Factoring trinomials of the form $ax^2 + bx + c$, where $a = 1$, resulted in the factored form $(x + g)(x + h)$, where $g \cdot h = c$ and $g + h = b$.

5. Use the examples in the table to explore factoring trinomials of the form $ax^2 + bx + c$ where $a \neq 1$. Summarize the similarities and differences of the factored forms where $a = 1$ and where $a \neq 1$.

Trinomial	$3x^2 + 17x + 10$	$3x^2 - 13x - 10$
Factored form	$(3x + 2)(x + 5)$	$(3x + 2)(x - 5)$
Similarities		
Differences		

When factoring quadratics of the form $ax^2 + bx + c$, the coefficient b is the sum of factors of the product of a and c .

6. In $3x^2 + 17x + 10$ and $3x^2 - 13x - 10$, the products of a and c are 30 and -30 , respectively.

a. List all the pairs of factors for both 30 and -30 .

Value	Pairs of Factors
30	
-30	

b. Circle the factor pair for 30 whose sum is 17.

c. Circle the factor pair for -30 whose sum is -13 .

d. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial expression.

$3x^2 + 17x + 10$	$3x^2 - 13x - 10$
$3x^2 + 2x + 15x + 10$	$3x^2 + 2x - 15x - 10$
$3x(x + \underline{\hspace{1cm}}) + 2(x + \underline{\hspace{1cm}})$	$3x(x - \underline{\hspace{1cm}}) + 2(x - \underline{\hspace{1cm}})$
$(3x + 2)(x + 5)$	$(3x + 2)(x - 5)$



7.4.3 Guided Practice: Factoring Trinomials

Where $a \neq 1$

1. Factor $5x^2 + 6x - 8$ by grouping.

2. Consider the trinomial $3x^2 - 13x - 10$.

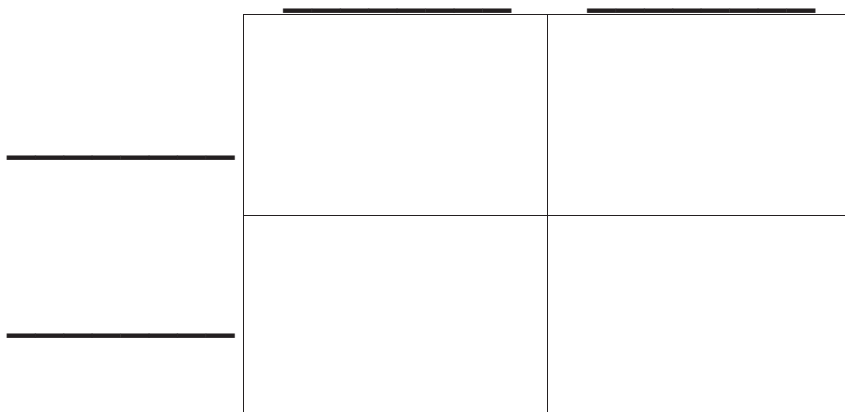
a. Complete the table by choosing the appropriate values.

	$3x$	☐ 1 ☐ -1 ☐ 2 ☐ -2 ☐ 5 ☐ -5 ☐ 10 ☐ -10
x	$3x^2$	☐ $1x$ ☐ $-1x$ ☐ $2x$ ☐ $-2x$ ☐ $5x$ ☐ $-5x$ ☐ $10x$ ☐ $-10x$
☐ 1 ☐ -1 ☐ 2 ☐ -2 ☐ 5 ☐ -5 ☐ 10 ☐ -10	☐ $3x$ ☐ $-3x$ ☐ $6x$ ☐ $-6x$ ☐ $15x$ ☐ $-15x$ ☐ $30x$ ☐ $-30x$	-10

b. Use the values to complete the area model.

	$3x$	
x	$3x^2$	
		-10

3. Factor $5x^2 - 3x - 8$ using the area model.



7.4.4 Try It: Factoring Trinomials Where $a \neq 1$

1. Factor $2z^2 - 21z - 36$.

2. Factor $3b^2 - 23b + 14$.

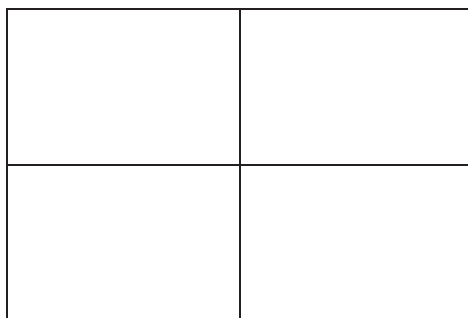


7.4.5 Your Turn!

1. Consider the quadratic expression.

$$5x^2 + 21x + 18$$

- a. Factor using the area model.



- b. Factor by grouping.



7.4.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on factoring trinomials where $a \neq 1$?

